

Drone-Remote Weapon Station Software - Architecture and Safety Classification (MIL-STD-882E)

1. Purpose

- This document explains the software architecture required for the Remote Weapon Station (RWS) and identifies which software elements are safety-critical versus non-safety software in accordance with MIL-STD-882E principles.

2. Software Classification per MIL-STD-882E

- MIL-STD-882E requires identification of software safety criticality and independence between safety authority and functional software.

3. Safety-Critical Software

- Independent Safety Monitor firmware – SAFE authority, watchdog, torque disable, fire inhibit
- Fire interlock logic – dual-channel firing authorization
- Drive enables gating logic – hardware/software interface ensuring torque-off on fault

4. Non-Safety (Functional) Software

- Operator GUI and HMI
- Video processing and streaming
- Aiming and stabilization algorithms
- Ballistic computation
- Tracking assist functions
- Data logging and telemetry display

5. Software Architecture Overview

- The RWS software is partitioned into Operator GUI, Fire Control Computer software, Independent Safety Monitor firmware, and Fire Interlock logic to ensure independence and fail-safe behavior.

6. Independence Principle

- Safety Monitor firmware operates independently of the Fire Control Computer to enforce SAFE state during comm loss, power fault, or software malfunction.

7. Safety Mechanisms

- Heartbeat supervision
- Dual-channel fire enables
- Hard torque disables
- Limit switch enforcement
- Watchdog timers

8. Verification Approach

- Verification includes fault injections, watchdog timeout testing, comm loss testing, and fire inhibit validation.

Software Safety Upgrade (MIL-STD-882E Extended)**9. Software Integrity Allocation**

Software Element	Integrity Level
ISM Firmware	High
Fire Interlock Logic	High
FCC Control Software	Medium
Operator GUI	Low

10. Software Safety Requirements Traceability

- Hazard → Safety Requirement → Software Implementation → Verification mapping established.

11. Software Independence Argument

- ISM firmware operates independently of FCC ensuring SAFE state despite FCC failure.

12. Software Safety Case Section

- Top claim: Software cannot produce hazardous firing without safety authorization.

13. Software Hazard Mapping

- Runaway motion → torque disable logic; unintended fire → interlock gating; comm loss → watchdog SAFE.

14. JSSEH Alignment

- Software lifecycle includes hazard analysis, traceability, verification, configuration control.

15. Cybersecurity Interface (RMF)

- Authentication, encrypted comm transport, anti-spoofing checks integrated.

16. Software Verification Matrix

- Includes HIL testing, fault injection, watchdog timeout validation, GUI usability validation.

17. Safety Software Package

- Software Architecture Diagram: FCC modules (comms, control, video, logging), ISM firmware safety core, interlock logic.
- FCC Software Decomposition: Comms service, turret control service, fire control service, health monitoring, logging.
- ISM Firmware State Machine: SAFE → ARMED → FIRING with fault transition to SAFE.
- GUI Safety Screen Specification: Clear SAFE/ARMED annunciation, dual confirmation firing, alarm prioritization.
- Message Protocol Tables: GCS↔FCC Ethernet messages; FCC↔ISM heartbeat; CAN motion and fault frames.
- Software FMEA: FCC crash mitigated by ISM SAFE; GUI error mitigated by interlock; comm corruption mitigated by CRC.
- Software Hazard Log Extension: Hazards mapped to software mitigations and verification evidence.
- Software Verification Procedures: HIL tests, fault injection, watchdog validation, endurance tests.
- Software SAR Evidence Section: Logs, fault reports, verification artifacts supporting safety approval.

UPGRADED DUAL-CHANNEL + ISM SAFETY ARCHITECTURE (INTEGRATED)

1. Safety Authority Architecture Upgrade

- The system is updated to include an Independent Safety Monitor (ISM) acting as the final safety authority.
- All actuator commands are validated and gated by the ISM prior to execution.

2. Safety Authority Separation (Critical Requirement)

- The ISM operates independently from the Fire Control Computer (FCC) and enforces a SAFE state regardless of FCC behavior, communication loss, or software malfunction.

3. Updated Software Architecture (Control vs Safety Separation)

- Control Computers generate motion commands, while ISM enforces safety constraints including torque disable, command validation, watchdog monitoring, and limit enforcement.

4. Hazard Mitigation Mapping

- Runaway motion → ISM torque disable
- Invalid command → ISM command validation
- Communication loss → ISM watchdog SAFE
- Timing fault → ISM supervision

5. FTA Contribution

- Top Event: Uncontrolled Motion
- Causes include ISM failure OR ISM pass-through of unsafe command.
- ISM is treated as the primary safety barrier.

6. Expanded FMECA – ISM Critical Failure Modes

- Failure Mode: ISM crash → Effect: Loss of safety authority → Severity: Catastrophic → Mitigation: Hardware fail-safe default OFF
- Failure Mode: ISM logic error → Effect: Unsafe command allowed → Mitigation: Minimal verified code
- Failure Mode: ISM comm loss → Effect: No supervision → Mitigation: Default SAFE
- Failure Mode: Watchdog failure → Effect: Fault undetected → Mitigation: Independent watchdog

7. STPA Control Structure

- Operator → FCC → ISM → Actuators
- Unsafe Control Actions include ISM allowing unsafe torque or failing to enforce SAFE.

8. Software Safety Classification

- ISM Firmware: High Integrity
- Interlock Logic: High
- FCC: Medium
- GUI: Low

9. Verification Enhancements

- Added ISM-specific tests:
 - Fault injection (invalid command)
 - Watchdog timeout
 - Communication loss
 - Corrupted data validation
 - Timing violation scenarios

10. Safety Case

- The system cannot produce hazardous motion because all actuator commands are gated by an independent safety authority (ISM) capable of enforcing a SAFE state under all failure conditions.

11. Software General Architecture - Expanded Engineering Data

Note: This expansion strengthens software architecture, safety classification, interfaces, verification evidence, and lifecycle controls.

12. Assumptions and System Boundaries

- The software boundary includes the Operator Control Unit (OCU), Fire/Function Control Computer (FCC), Independent Safety Monitor (ISM), motor-drive interface, sensor interface, video pipeline, logging service, and maintenance interface.
- The ISM is treated as the final safety authority for motion-enable, torque-disable, command validity, heartbeat supervision, and transition to SAFE state.
- Functional software may request motion or mode changes, but it cannot directly energize actuators or bypass safety gates.
- Any fire-control or payload-control functions remain subordinate to independent interlocks, positive operator intent, and system SAFE/ARMED/FIRING state logic.
- The architecture assumes no autonomous engagement authority. Operator authorization and independent safety gating are required for hazardous actions.

13. Expanded Logical Software Architecture

Layer / Component	Primary Responsibility	Safety Role	Typical Evidence
Operator GUI / HMI	Displays status, video, alarms, commands, and operator confirmations.	Low safety authority; must present clear state and alarms without granting authority.	HMI requirements, usability review, alarm-priority test records.
Communications Service	Manages network sessions, message parsing, authentication status, heartbeat forwarding, and link health.	Supports safety by reporting link state; does not make final safety decisions.	Protocol tests, malformed-message tests, timeout tests.
FCC Control Services	Generates requested pan/tilt motion, stabilization requests, health reports, and non-safety payload commands.	Medium integrity; commands are advisory until validated by ISM.	Unit tests, integration tests, HIL command-response logs.
Video / Sensor Services	Handles camera selection, video encoding/decoding, timestamping, and sensor acquisition.	Non-safety; must not block safety monitor or torque-off path.	Latency measurements, failure-mode tests, resource load tests.
Independent Safety Monitor Firmware	Validates commands, enforces limits, supervises heartbeat, controls drive-enable and safe state.	High-integrity safety authority.	Code review, static analysis, fault injection, watchdog and safe-state verification.
Fire / Hazardous-Action Interlock Logic	Implements dual-channel authorization and inhibit gating for hazardous actions.	High-integrity final interlock barrier.	Dual-channel verification, inhibit tests, independence evidence.
Motor Drive / I/O Interface	Executes allowed motion only when enable conditions are satisfied.	Fail-safe output stage; defaults to disabled on fault.	Torque-disable tests, limit enforcement tests, power-fault tests.
Data Logger / BIT Service	Records operational, maintenance, and safety events.	Safety evidence support; must preserve event sequence and fault context.	Log integrity tests, timestamp checks, post-fault replay review.

14. Process and Task Partitioning

Task / Process	Suggested Priority	Execution Rate / Trigger	Isolation Requirement	Failure Behavior
ISM safety loop	Highest / deterministic	Fixed cyclic loop plus interrupt fault handling	Dedicated firmware task or separate MCU where practical	Immediate SAFE and torque disable on fault.
Drive command validation	High	Every command before drive-enable update	Separated from FCC command generation	Reject invalid command and log fault.
FCC control loop	High	Deterministic loop sized to actuator dynamics	Cannot write final enable directly	Hold last safe request or request SAFE on invalid state.
OCU command receiver	Medium	On message arrival plus heartbeat timer	Input validation before forwarding	Drop invalid packet and flag comm fault.
Video pipeline	Medium / best effort	Frame-based	Resource-limited so it cannot starve safety loop	Drop frames rather than delay safety functions.
Telemetry and logging	Low to medium	Periodic and event-driven	Non-blocking queue recommended	Preserve critical fault record; degrade gracefully if storage full.
Maintenance diagnostics	Low, controlled mode only	On authorized maintenance session	Disabled in operational mode where hazardous effects are possible	Terminate session and revert SAFE on authentication or integrity fault.

15. Safety State Machine Definition

- **SAFE:** Default state after power-up, reset, communication loss, watchdog timeout, invalid command, sensor fault, limit violation, or ISM internal fault. Actuator torque and hazardous-action outputs remain inhibited.
- **STANDBY:** System health is acceptable and communication is present, but hazardous outputs remain inhibited. Motion may remain disabled or limited depending on system design authority.
- **ARMED:** System has passed required health checks and operator authorization sequence. ISM continues validating all commands and may return to SAFE at any time.
- **MOTION ENABLED:** Drive-enable is permitted only while command, limits, heartbeat, power, and sensor plausibility remain valid.
- **FAULT:** Latched or non-latched condition requiring SAFE transition, event logging, operator annunciation, and defined recovery action.
- **MAINTENANCE:** Restricted mode for calibration, built-in-test, log download, and software update. Hazardous outputs are inhibited unless an approved maintenance safety procedure explicitly permits controlled testing.

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Transition	Required Condition	ISM Action	Evidence / Test
Power-up -> SAFE	Reset or cold start	Initialize outputs disabled; run BIT	Power-up safe-state test.
SAFE -> STANDBY	BIT pass, valid configuration, valid communication	Allow status reporting only	BIT and configuration validation.
STANDBY -> ARMED	Positive operator action plus valid health state	Enable armed annunciation; keep interlocks active	HMI confirmation and interlock test.
ARMED -> MOTION ENABLED	Valid motion request, valid limits, heartbeat in range	Enable drive gate for validated command only	Motion command boundary test.
Any State -> SAFE	Comm loss, watchdog timeout, invalid command, power fault, limit fault, ISM fault	Disable torque / hazardous output, log event, annunciate alarm	Fault-injection and timeout tests.
FAULT -> SAFE / STANDBY	Fault cleared and reset policy satisfied	Remain SAFE until authorized reset	Fault recovery test and log review.

16.Interface Control Data

Interface	Data Exchanged	Integrity / Safety Controls	Recommended Failure Response
OCU <-> FCC Ethernet	Commands, video control, telemetry, alarms, system mode requests.	Authentication, session control, sequence counters, CRC/message authentication, heartbeat.	Reject message, alarm, enter SAFE on timeout or repeated integrity failures.
FCC <-> ISM Safety Link	Heartbeat, requested command envelope, mode request, health summary, ISM status.	Independent heartbeat supervision, command range checks, monotonic counters, fail-safe default.	ISM disables outputs and reports SAFE/fault.
FCC/ISM <-> Motor Drives	Validated motion commands, enable line, fault line, limit information.	Separate enable path, hard torque-disable, drive-fault monitoring, limit enforcement.	Torque off and latched fault if unsafe.
Sensor Bus	IMU, encoder, limit switches, temperature, voltage, current, optional position feedback.	Plausibility checks, stale-data detection, range limits, redundant critical sensors where required.	Reject command or derate/disable motion.
Maintenance Port	Configuration, software update package, diagnostic reports, log export.	Role-based access, signed update package, audit trail, maintenance-mode lockout.	Abort session, preserve current safe state, log security event.

17.Message and Data Integrity Rules

- Every safety-relevant message should include message type, source, destination, sequence counter, timestamp or age indicator, payload length, payload, and integrity check.
- Commands are accepted only if they are fresh, authenticated, correctly sequenced, within allowed range, compatible with current state, and consistent with physical limits.
- Default behavior for missing, stale, malformed, replayed, out-of-range, or contradictory messages is rejection and transition to SAFE when the condition affects safety.
- The safety monitor should never rely on GUI state alone. It must use independent input validation and local safety state.
- Safety-related logs should include timestamp, source, command/request, decision, reason for rejection, state before/after, and fault code.

18. Software Safety Requirements - Expanded Traceability Set

Req. ID	Requirement	Allocated To	Verification Method
SWR-SAFE-001	The system shall enter SAFE state on power-up, reset, watchdog timeout, communication loss, invalid command, or safety-monitor internal fault.	ISM / Interlock	Fault injection, reset test, comm-loss test.
SWR-SAFE-002	The FCC shall not directly energize actuator torque or hazardous-action outputs without ISM validation and enable gating.	FCC / ISM / I/O	Architecture review, interface test, code review.
SWR-SAFE-003	The ISM shall validate command range, rate, direction, current state, and limit status before enabling motion.	ISM	Boundary-value tests, HIL tests.
SWR-SAFE-004	All hazardous-action commands shall require independent interlock authorization and clear operator state annunciation.	Interlock / HMI	Interlock test, HMI review, misuse-case test.
SWR-SAFE-005	Safety-critical software shall be separated from non-safety services so that video, logging, or GUI failure cannot block torque disable.	System Architecture	Design review, resource exhaustion test.
SWR-SAFE-006	Safety-relevant faults shall be logged with time, state, source, and mitigation outcome.	ISM / Logger	Log review after injected faults.
SWR-SAFE-007	Software update and configuration loading shall verify authenticity and integrity before activation.	Bootloader / Maintenance	Signed-package test, rollback test.
SWR-SAFE-008	The system shall provide a controlled recovery path from FAULT that requires fault clearance and authorized reset.	ISM / HMI	Recovery procedure test.

19. Hazard-to-Software Control Mapping – Extended in ANNEX A

Hazard	Software Control	Independent Barrier	Verification Evidence
Runaway pan/tilt motion	Range/rate validation, heartbeat supervision, torque-disable command.	ISM plus hard enable line.	Runaway command injection and torque-off timing test.
Unexpected motion after comm loss	Communication watchdog and SAFE transition.	ISM local timer independent of FCC.	Cable-pull and network-loss test.
Limit overtravel	Limit-switch enforcement and position plausibility checks.	Hardware limit plus ISM limit logic.	Limit actuation test and encoder fault test.
Corrupted or replayed command	Sequence counter, freshness, integrity check, authentication.	ISM rejection logic.	Malformed/replay packet test.
Unintended hazardous action	Dual-channel authorization, interlock gating, armed-state validation.	Independent interlock circuit / logic.	Inhibit validation and dual-channel mismatch test.
Loss of safety monitor	Fail-safe hardware default OFF and independent watchdog.	External watchdog / power gating.	ISM crash and watchdog test.
Misleading operator display	Alarm priority, SAFE/ARMED state annunciation, stale-data indication.	ISM status as authoritative source.	HMI usability and stale data test.

20. Cybersecurity and Safety Interface Requirements

- Cybersecurity controls should support safety by preventing unauthorized command injection, spoofed state transitions, replay attacks, unauthorized configuration changes, and unsafe software updates.
- Safety functions must remain deterministic under cyber stress. High traffic, malformed packets, or failed authentication must not delay torque-disable or SAFE transitions.
- Maintenance interfaces should be disabled or restricted during operational modes and should require explicit authorization, logging, and configuration-control records.
- Software update packages should be signed, versioned, rollback-capable, and tested before field activation.
- Security events that affect command validity or system state should be logged as safety-relevant events.

Threat / Condition	Control	Safety Effect
Command spoofing	Authentication, integrity check, source validation.	Prevents unauthorized state or motion request acceptance.
Replay message	Sequence counter, timestamp/age check.	Rejects stale commands and prevents unintended repeated actions.
Malformed packet	Strict parser, length check, safe rejection.	Avoids parser crash or unsafe default.
Denial of service	Rate limiting, safety-loop priority, independent ISM.	Safety monitor remains able to enter SAFE.
Unauthorized update	Signed package, role control, audit trail.	Prevents unverified software from controlling safety-relevant behavior.

21. Built-In Test (BIT), Health Monitoring, and Diagnostics

- Power-On BIT should verify memory integrity, firmware version, configuration validity, watchdog status, safety I/O default state, sensor presence, and communication readiness.
- Continuous BIT should monitor heartbeat, command validity, encoder plausibility, limit inputs, drive faults, supply voltage, temperature, CPU load, memory health, and queue overruns.
- Initiated BIT should be available in maintenance mode for deeper tests such as sensor calibration checks, drive-interface checks, safe-output tests, and log export.
- Fault reporting should separate warning, degraded, fault, and latched-fault categories. Latched faults require authorized reset after the unsafe condition is removed.

22.Configuration Management and Software Lifecycle Controls

Lifecycle Item	Control Requirement	Evidence
Requirements baseline	Each safety requirement has a unique ID and verification method.	Requirements traceability matrix.
Design baseline	Architecture, interface control, and safety partitioning are version controlled.	Approved design package.
Source code	Peer review and static analysis for safety-critical components.	Review records, analysis reports.
Build process	Reproducible build with compiler/toolchain version capture.	Build log, checksum, release tag.
Configuration data	Range limits, calibration, and mode settings are controlled and validated.	Configuration file signature and change log.
Release approval	Safety-impact assessment before software deployment.	Release note, hazard-log update, regression test report.
Field feedback	Fault logs and anomaly reports feed back into hazard analysis.	Problem report, corrective-action record.

23.Verification and Validation Matrix - Expanded

Test ID	Objective	Method	Pass Criteria
V&V-001	Verify SAFE default on power-up and reset.	Power-cycle and reset injection.	Outputs remain disabled until authorized state transition.
V&V-002	Verify communication-loss transition.	Disconnect OCU/FCC link during operation.	ISM transitions to SAFE within defined timeout and logs event.
V&V-003	Verify invalid command rejection.	Inject out-of-range, stale, replayed, and malformed messages.	Command rejected; no unsafe output enabled.
V&V-004	Verify watchdog behavior.	Stop FCC heartbeat and simulate ISM task hang.	Appropriate watchdog forces SAFE / hardware OFF.
V&V-005	Verify limit enforcement.	Command motion toward simulated/actual limit.	Motion inhibited or stopped at limit with fault record.
V&V-006	Verify video/logging failure isolation.	Overload video stream and storage.	Safety loop still executes and torque-disable remains available.
V&V-007	Verify interlock independence.	Attempt hazardous action without valid dual-channel authorization.	Action remains inhibited and alarm/log are generated.
V&V-008	Verify cyber-stress safety.	Send malformed/high-rate unauthenticated traffic.	System rejects traffic without safety-function starvation.
V&V-009	Verify software update controls.	Load unsigned, corrupted, rollback, and valid packages.	Only valid authorized package activates; failures leave prior safe version.
V&V-010	Verify fault recovery.	Clear fault and perform authorized reset.	System remains SAFE until reset criteria are satisfied.

24. Software FMEA Additions

Software Item	Failure Mode	Effect	Mitigation / Detection
GUI / HMI	Displays stale SAFE/ARMED state.	Operator may misunderstand system state.	State timestamp, stale-data alarm, ISM-authoritative status.
Communications parser	Crash or accepts malformed message.	Loss of control or unsafe request forwarding.	Strict parser, fuzz testing, safe rejection, watchdog.
FCC motion service	Generates excessive rate or invalid position request.	Potential unsafe motion request.	ISM range/rate validation and torque gate.
Video service	CPU/resource starvation.	Delayed functional response.	Safety loop priority and resource partitioning.
Logger	Storage full or corrupted log.	Reduced investigation evidence.	Non-blocking logging, critical event reservation, health alarm.
ISM firmware	Logic error permits unsafe command.	Loss of primary safety barrier.	Minimal verified code, requirements traceability, independent review, HIL tests.
Bootloader/update	Loads unauthorized firmware.	Compromised safety behavior.	Signed updates, secure version control, rollback and validation.

25. Documentation Package for Safety Review

- Software Architecture Description with partitioning and data-flow diagrams.
- Interface Control Document (ICD) for OCU-FCC, FCC-ISM, drives, sensors, and maintenance interfaces.
- Software Safety Requirements Specification and traceability matrix.
- Software Hazard Analysis, FMEA/FMECA, FTA contribution, and STPA worksheet.
- Software Configuration Management Plan and secure update procedure.
- Verification Procedures, test reports, fault-injection logs, and HIL evidence.
- Safety Case with claims, arguments, evidence, assumptions, and open issues.
- Cybersecurity risk assessment focused on safety-relevant command and update paths.
- Operator HMI specification including alarms, mode annunciation, and fault recovery behavior.

26. Open Engineering Decisions / Data Needed

Topic	Data Needed	Reason
Safety timing budget	Maximum allowed time from detected fault to torque-disable / SAFE output.	Required for watchdog and fault-response verification.
Command envelope	Approved pan/tilt position, speed, acceleration, and rate limits.	Required for ISM validation rules.
Sensor redundancy	Which safety-relevant signals are single-channel versus dual-channel.	Required for safety integrity argument.
Fault reset policy	Which faults are latched and what authorization is required for reset.	Required for HMI and safety-state behavior.
Cybersecurity baseline	Authentication, encryption, update-signing, and logging policies.	Required for RMF/security-safety interface evidence.
Environmental assumptions	Temperature, vibration, power quality, EMI/EMC limits.	Required for BIT thresholds and qualification tests.
Software toolchain	Compiler, static analysis, test framework, build server, release process.	Required for configuration-control evidence.

Hazard Log Extension

Software architecture derived from the 40-hazard preliminary RAC log

27.Purpose of Hazard Log Integration

This section extends the software architecture using the hazard log as the driving safety input. The intent is to convert hazard-log mitigations into software safety requirements, architecture controls, interface rules, verification evidence, and safety-case arguments. The extension does not grant functional software direct authority over hazardous outputs; those outputs remain subordinate to the Independent Safety Monitor (ISM), interlock logic, positive operator intent, and fail-safe hardware behavior.

- Hazard-log mitigations are allocated to software only where software can detect, inhibit, annunciate, log, or enforce a safe-state transition.
- Hardware and procedural mitigations remain credited as independent barriers and are not replaced by software controls.
- Each software control requires verification evidence, including fault injection, HIL testing, cybersecurity tests, HMI validation, or recovery testing as applicable.
- The software safety case should reference the hazard ID, associated software requirement, verification artifact, residual risk, and any open assumptions.

28.Hazard Log Summary and Software-Relevance Assessment

Category	Count	Software Safety Interpretation
Total hazards reviewed	40	All 40 hazard-log rows are considered for software traceability.
RAC 1 (High)	4	Requires explicit mitigation/evidence path where software participates in detection, inhibition, or annunciation.
RAC 2 (Serious)	24	Requires explicit mitigation/evidence path where software participates in detection, inhibition, or annunciation.
RAC 3 (Medium)	10	Requires explicit mitigation/evidence path where software participates in detection, inhibition, or annunciation.
RAC 4 (Low)	2	Requires explicit mitigation/evidence path where software participates in detection, inhibition, or annunciation.
Software-relevant hazards	40	Every listed hazard has at least one software-related monitoring, indication, logging, command-validation, or inhibit requirement.

29.Hazard-Derived Software Requirement Families

Req. Family	Requirement	Hazard Coverage	Primary Verification
SWR-HZ-FIRE-01	For any armed or hazardous-action state, the software shall require valid state, operator confirmation, interlock authorization, current health status, and no active inhibits before passing the request to the independent inhibit path.	H1-H5, H8, H16-H17, H32-H33, H39-H40	Interlock tests, dual-channel mismatch tests, HMI misuse-case tests, fault injection.
SWR-HZ-MOTION-01	The ISM shall reject motion requests that violate allowed position, rate, acceleration, limit, homing, power, encoder, or platform-stability constraints.	H6-H7, H18, H27, H36, H38	HIL envelope tests, induced encoder faults, limit-switch tests, torque-disable timing.
SWR-HZ-COMMS-01	Safety-relevant messages shall be authenticated, sequenced, freshness-checked, and timeout-supervised; missing,	H8-H10, H29-H31, H40	Network emulation, replay/spoof tests, malformed-packet tests, latency and congestion testing.

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	stale, malformed, replayed, or unauthorized messages shall be rejected and escalated to SAFE where required.		
SWR-HZ-HEALTH-01	The software shall monitor thermal, current, voltage, stability, actuator, and critical sensor health and shall derate, inhibit, or transition to SAFE when thresholds are exceeded.	H11-H14, H20, H36, H38	Threshold injection, endurance tests, thermal/current monitoring validation, flight or platform integration tests.
SWR-HZ-SENSOR-01	The software shall detect stale, invalid, uncalibrated, or low-confidence sensor/video/aim-assist data and prevent use of such data for hazardous state transitions.	H21-H26, H39-H40	Sensor freeze, stale-frame, confidence, calibration, zeroing/configuration, and HMI warning tests.
SWR-HZ-MAINT-01	Maintenance, transport, inspection, configuration, and diagnostic modes shall maintain lockout of hazardous outputs unless explicitly authorized under approved safe-test conditions.	H13, H15, H28, H33-H35	Maintenance-mode tests, lockout/tagout verification, BIT checks, checklist prompts, log review.
SWR-HZ-RECOVERY-01	After power loss, reboot, software crash, or latched fault, the system shall restart in SAFE and require health checks and authorized reset/re-authorization before operational states can resume.	H18-H19, H30	Power-cycle tests, crash/reboot fault injection, recovery procedure validation.

30. Hazard-Driven Architecture Updates

Architecture Element	Update Required by Hazard Log	Hazard-Reduction Rationale
ISM safety loop	Final authority for SAFE, motion enable, command acceptance, watchdog supervision, and fault-state transition.	Supports hazards involving software faults, comm loss, runaway motion, power/reboot anomalies, and unsafe state transitions.
Interlock logic	Independent hazardous-action inhibit barrier; receives only validated requests and has feedback/diagnostic checks.	Supports unintended firing, stuck output, armed/disarm faults, operator-error, and blind-operation hazards.
FCC services	Functional command generation only; no direct write access to final enable or inhibit outputs.	Prevents GUI, video, tracking, or control-loop failure from becoming sole cause of hazardous output.
HMI and OCU	Clear SAFE/ARMED/FAULT annunciation, warning priority, stale-data indicators, guarded mode changes, and guided recovery prompts.	Addresses operator error, mode confusion, loss of video, target-designation confusion, and maintenance hazards.
Communications layer	Authentication, integrity checks, sequence counters, heartbeat, latency measurement, anti-replay, and fail-closed parsing.	Addresses spoofing, replay, dropped packets, latency, EMI/control upset, and link-loss hazards.
Health and BIT service	Power-on, continuous, and initiated BIT with fault logging and degraded/latched-fault categories.	Addresses sensor failures, thermal/current/power faults, structural/maintenance cues, and post-event evidence needs.

31.Expanded Hazard-to-Software Control Matrix

The table below links each preliminary hazard-log entry to software domains, hazard-derived requirement families, specific software controls, and verification evidence. Hardware and procedural controls from the hazard log remain independent barriers where applicable.

HZ ID	Hazard / RAC	Software Domain	Allocated SW Req.	Concise Software Safety Control	Verification / Evidence
H1	Unintended firing (SW logic fault) RAC 2 (Serious)	Fire/hazardous-action interlock; Comms integrity and cybersecurity	SWR-HZ-FIRE-01	state/interlock inhibit.	SW unit/integration tests; SW: interlock/inhibit.
H2	Unintended firing (HW output stuck ON) RAC 2 (Serious)	Fire/hazardous-action interlock; Power/thermal health monitoring; Maintenance, mechanical, and BIT interfaces	SWR-HZ-FIRE-01	state/interlock inhibit.	Bench electrical fault tests (short/open); SW: interlock/inhibit.
H3	Unintended firing (operator error) RAC 1 (High)	Fire/hazardous-action interlock; Motion control and ISM gating; Comms integrity and cybersecurity	SWR-HZ-FIRE-01	state/interlock inhibit.	Human factors review; SW: interlock/inhibit.
H4	Unintended firing (spurious EM/ESD) RAC 2 (Serious)	Fire/hazardous-action interlock; Comms integrity and cybersecurity	SWR-HZ-FIRE-01	state/interlock inhibit.	EMI/ESD tests (MIL-STD-461 as applicable); SW: interlock/inhibit.
H5	Firing while turret not in authorized sector RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Comms integrity and cybersecurity	SWR-HZ-FIRE-01	state/interlock inhibit.	Range test; SW: interlock/inhibit.
H6	Turret runaway (yaw) RAC 3 (Medium)	Motion control and ISM gating; Sensor/video/aim-assist safeguards	SWR-HZ-MOTION-01	motion envelope + torque-off.	HIL stability tests; SW: HIL envelope/torque-off.
H7	Turret runaway (pitch) RAC 3 (Medium)	Motion control and ISM gating; Sensor/video/aim-assist safeguards	SWR-HZ-MOTION-01	motion envelope + torque-off.	Motion tests at extremes; SW: HIL envelope/torque-off.
H8	Loss of comms while armed RAC 1 (High)	Fire/hazardous-action interlock; Comms integrity and cybersecurity; Maintenance, mechanical, and BIT interfaces	SWR-HZ-FIRE-01, SWR-HZ-COMMS-01	state/interlock inhibit; heartbeat/auth/freshness checks.	Comm-loss tests (RF off/jam); SW: network/cyber/latency.
H9	Comm spoofing / unauthorized control RAC 2 (Serious)	Comms integrity and cybersecurity	SWR-HZ-COMMS-01	heartbeat/auth/freshness checks.	Pen-test; SW: network/cyber/latency.
H10	Latency / dropped packets causing wrong aim at FIRE moment RAC 2 (Serious)	Fire/hazardous-action interlock; Comms integrity and cybersecurity	SWR-HZ-COMMS-01	heartbeat/auth/freshness checks.	Network emulation tests; SW: network/cyber/latency.
H11	Stabilization failure during firing RAC 2 (Serious)	Fire/hazardous-action interlock; Sensor/video/aim-assist safeguards	SWR-HZ-HEALTH-01	health thresholds + derate/inhibit.	Firing tests; SW: threshold injection.
H12	Recoil mitigation failure (mechanical) RAC 3 (Medium)	Fire/hazardous-action interlock; Motion control and ISM gating	SWR-HZ-HEALTH-01	health thresholds + derate/inhibit.	Endurance firing test; SW: threshold injection.
H13	Mount structural failure RAC 2 (Serious)	Motion control and ISM gating; Maintenance, mechanical, and BIT interfaces	SWR-HZ-MAINT-01	BIT/service lockout + logs.	Static load + fatigue test; SW: BIT/lockout/checklist.
H14	Weapon/munition overheating (cook-off) RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Sensor/video/aim-assist safeguards; Power/thermal health monitoring	SWR-HZ-HEALTH-01	health thresholds + derate/inhibit.	Thermal tests; SW: threshold injection.
H15	Ammunition box / feed chute jam RAC 3 (Medium)	Fire/hazardous-action interlock; Maintenance, mechanical, and BIT interfaces	SWR-HZ-MAINT-01	BIT/service lockout + logs.	Jam tests; SW: BIT/lockout/checklist.
H16	Runaway burst (trigger stuck) RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Comms integrity and cybersecurity; Maintenance, mechanical, and BIT interfaces	SWR-HZ-FIRE-01	state/interlock inhibit.	Burst-limit test; SW: interlock/inhibit.
H17	Failure to DISARM (arm latch stuck) RAC 3 (Medium)	Fire/hazardous-action interlock; Maintenance, mechanical, and BIT interfaces	SWR-HZ-FIRE-01	state/interlock inhibit.	Fault-injection; SW: interlock/inhibit.

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H# ID	Hazard / RAC	Software Domain	Allocated SW Req.	Concise Software Safety Control	Verification / Evidence
H18	Unexpected motion at power-up RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Power/thermal health monitoring	SWR-HZ-MOTION-01, SWR-HZ-RECOVERY-01	motion envelope + torque-off; SAFE restart + authorized recovery.	Power-cycle tests; SW: HIL envelope/torque-off, reboot/recovery.
H19	Loss of power during armed state RAC 2 (Serious)	Fire/hazardous-action interlock; Comms integrity and cybersecurity; Power/thermal health monitoring	SWR-HZ-RECOVERY-01	SAFE restart + authorized recovery.	Power interruption tests; SW: reboot/recovery.
H20	Battery thermal runaway / fire RAC 2 (Serious)	Fire/hazardous-action interlock; Power/thermal health monitoring	SWR-HZ-HEALTH-01	health thresholds + derate/inhibit.	Abuse tests (as applicable); SW: threshold injection.
H21	Laser pointer hazard (eye exposure) RAC 2 (Serious)	Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Laser safety test; SW: stale/confidence/HMI.
H22	Rangefinder emits when not intended RAC 3 (Medium)	Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Functional tests; SW: stale/confidence/HMI.
H23	Sensor failure leads to incorrect aim indication RAC 2 (Serious)	Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Sensor fault injection; SW: stale/confidence/HMI.
H24	Auto-tracking drifts / mis-tracks RAC 2 (Serious)	Fire/hazardous-action interlock; Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Tracking evaluation; SW: stale/confidence/HMI.
H25	Ballistic solution wrong RAC 3 (Medium)	Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Range zero tests; SW: stale/confidence/HMI.
H26	Ammunition count error RAC 4 (Low)	Fire/hazardous-action interlock; Sensor/video/aim-assist safeguards	SWR-HZ-SENSOR-01	sensor freshness/confidence gating.	Counter validation; SW: stale/confidence/HMI.
H27	Mechanical stop failure RAC 3 (Medium)	Motion control and ISM gating	SWR-HZ-MOTION-01	motion envelope + torque-off.	End-stop overload test; SW: HIL envelope/torque-off.
H28	Cable snag/entanglement with DRONE RAC 3 (Medium)	Motion control and ISM gating	SWR-HZ-MAINT-01	BIT/service lockout + logs.	Fit-check; SW: BIT/lockout/checklist.
H29	EMI causes control upset RAC 3 (Medium)	System health monitoring and event logging	SWR-HZ-COMMS-01	heartbeat/auth/freshness checks.	EMI tests; SW: network/cyber/latency.
H30	Software crash / reboot during operation RAC 2 (Serious)	System health monitoring and event logging	SWR-HZ-COMMS-01, SWR-HZ-RECOVERY-01	heartbeat/auth/freshness checks; SAFE restart + authorized recovery.	Stress tests; SW: network/cyber/latency, reboot/recovery.
H31	Clock/time sync error affects command validity RAC 4 (Low)	Fire/hazardous-action interlock; Comms integrity and cybersecurity	SWR-HZ-COMMS-01	heartbeat/auth/freshness checks.	Time-drift tests; SW: network/cyber/latency.
H32	Inadvertent mode change (SAFE/ARM/AUTO) RAC 2 (Serious)	Fire/hazardous-action interlock	SWR-HZ-FIRE-01	state/interlock inhibit.	Human factors test; SW: .
H33	Maintenance hazard – arming during service RAC 1 (High)	Fire/hazardous-action interlock; Maintenance, mechanical, and BIT interfaces	SWR-HZ-FIRE-01, SWR-HZ-MAINT-01	state/interlock inhibit; BIT/service lockout + logs.	Maintenance procedure audit; SW: BIT/lockout/checklist.
H34	Transport hazard – accidental actuation RAC 2 (Serious)	Maintenance, mechanical, and BIT interfaces	SWR-HZ-MAINT-01	BIT/service lockout + logs.	Transport vibration test; SW: BIT/lockout/checklist.
H35	Weapon release/quick disconnect failure RAC 2 (Serious)	Maintenance, mechanical, and BIT interfaces	SWR-HZ-MAINT-01	BIT/service lockout + logs.	Drop/load test; SW: BIT/lockout/checklist.
H36	Overcurrent in actuator leads to fire RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Power/thermal health monitoring	SWR-HZ-MOTION-01, SWR-HZ-HEALTH-01	motion envelope + torque-off; health thresholds + derate/inhibit.	Stall tests; SW: HIL envelope/torque-off, threshold injection.
H37	Shot counter / feedback sensor failure hides firing RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating; Sensor/video/aim-assist safeguards; Maintenance, mechanical, and BIT interfaces	SWR-SAFE-001	fault log + SAFE response.	Sensor disconnect tests; SW: interlock/inhibit.
H38	DRONE flight controller interaction causes oscillations RAC 2 (Serious)	Fire/hazardous-action interlock; Motion control and ISM gating	SWR-HZ-MOTION-01, SWR-HZ-HEALTH-01	motion envelope + torque-off; health thresholds + derate/inhibit.	Flight tests; SW: HIL envelope/torque-off, threshold injection.
H39	Friendly fire due to wrong target designation RAC 1 (High)	Fire/hazardous-action interlock; Comms integrity and cybersecurity; Sensor/video/aim-assist safeguards	SWR-HZ-FIRE-01, SWR-HZ-SENSOR-01	state/interlock inhibit; sensor freshness/confidence gating.	Operational scenario tests; SW: stale/confidence/HMI.
H40	Loss of video feed while armed RAC 2 (Serious)	Fire/hazardous-action interlock; Comms integrity and cybersecurity; Sensor/video/aim-	SWR-HZ-FIRE-01, SWR-HZ-COMMS-01, SWR-HZ-SENSOR-01	state/interlock inhibit; heartbeat/auth/freshness checks; sensor freshness/confidence gating.	Video loss tests; SW: network/cyber/latency, stale/confidence/HMI.

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Hz ID	Hazard / RAC	Software Domain	Allocated SW Req.	Concise Software Safety Control	Verification / Evidence
		assist safeguards; Maintenance, mechanical, and BIT interfaces			

32. Hazard-Driven Verification Additions

Test ID	Objective	Method	Pass Criteria
V&V-HZ-001	Hazard-log regression suite	Run all tests mapped in the hazard-to-software matrix after each safety-impacting release.	Every hazard has a passed, failed, or waived test result with justification and reviewer approval.
V&V-HZ-002	Unsafe-state transition tests	Inject comm loss, watchdog timeout, invalid command, power interruption, sensor stale/fault, and limit violation.	System transitions to SAFE or inhibited state within the approved timing budget and records the event.
V&V-HZ-003	Cyber-safety tests	Run spoof, replay, malformed packet, high-rate traffic, and unauthorized update attempts.	Unauthorized or invalid data is rejected; safety loop is not starved; no direct hazardous output is enabled.
V&V-HZ-004	HMI misuse-case validation	Evaluate guarded controls, state annunciation, warnings, stale-data indicators, mode changes, and recovery prompts with representative users.	Operators can correctly identify SAFE/ARMED/FAULT state and cannot trigger hazardous transitions through single accidental action.
V&V-HZ-005	Health-monitor threshold validation	Inject or simulate thermal, current, voltage, stability, IMU/encoder, and video/sensor anomalies.	Software detects threshold crossings, derates or inhibits as specified, logs the fault, and preserves recovery constraints.
V&V-HZ-006	Maintenance and service-mode validation	Verify lockout, role authorization, physical/procedural checklist prompts, safe update, and log export behavior.	Hazardous outputs remain inhibited in service states unless a separately approved safe-test procedure is active.

33. Required Evidence Artifacts for Safety Review

- Hazard-to-software requirements traceability matrix showing H1-H40 coverage by requirement, design control, code module, and verification procedure.
- ISM and interlock verification report, including safe-state transition timing, watchdog tests, command rejection tests, and fault-injection logs.
- Communications and cybersecurity safety test report covering authentication, command integrity, anti-replay, latency, dropped packets, malformed packets, and denial-of-service stress.
- HMI human-factors evidence for SAFE/ARMED/FAULT annunciation, mode-change confirmation, warning priority, stale-video indication, and recovery guidance.
- Health-monitor and BIT evidence for sensors, power, thermal, current, video, actuator, and storage/logging fault cases.
- Configuration-management evidence showing controlled command envelopes, limits, calibration data, software versions, signed update packages, rollback behavior, and release approval.

34. Safety Case Updates Driven by Hazard Log

Safety Case Claim	Claim Statement	Evidence / Argument
Claim SC-HZ-1	No single FCC, GUI, video, tracking, or communications software fault can directly enable a hazardous output.	Architecture partitioning; ISM final authority; interlock independence; fault-injection and interface tests.
Claim SC-HZ-2	The system transitions to SAFE or inhibited state for loss of communication, invalid command, watchdog timeout, power/reboot anomaly, or critical sensor/limit fault.	SWR-HZ-COMMS-01, SWR-HZ-MOTION-01, SWR-HZ-RECOVERY-01, V&V-HZ-002 evidence.
Claim SC-HZ-3	Operator-facing software reduces mode confusion and blind-operation hazards through clear annunciation, guarded controls, and stale-data indications.	HMI specification; usability tests; UI simulation; HMI misuse-case validation.
Claim SC-HZ-4	Cybersecurity controls support safety by preventing unauthorized, replayed, malformed, or stale commands from being accepted as valid control inputs.	Authentication/design review; replay/spoof testing; malformed packet tests; secure update tests.
Claim SC-HZ-5	Maintenance and recovery functions cannot bypass safety states without explicit approved controls and auditable evidence.	Maintenance lockout tests; signed update procedure; authorized reset logs; service-mode review.

35.35. Open Data Required to Close Hazard-Derived Software Items

Open Item	Data Needed	Why It Matters
Fault response timing budget	Maximum time allowed from detection to torque-disable/SAFE for comm loss, watchdog timeout, invalid command, limit violation, and power fault.	Needed to set watchdog periods, heartbeat timeout, and pass/fail criteria.
Approved command envelope	Pan/tilt position, rate, acceleration, current, and platform attitude limits, including reduced envelopes for degraded states.	Needed for ISM validation and HIL boundary tests.
Sensor freshness and confidence thresholds	Maximum age for video frames, IMU/encoder data, target/designation information, and tracking confidence.	Needed for stale-data inhibit rules and HMI warnings.
Latched fault list and reset authority	Which faults require maintenance reset, operator reset, power cycle, or service inspection.	Needed for FAULT-to-SAFE/STANDBY recovery logic.
Cybersecurity baseline	Approved authentication, encryption, key management, anti-replay, secure boot, signed-update, and audit policies.	Needed for RMF/security-safety evidence and protocol testing.
Environmental qualification assumptions	EMI/EMC, vibration, temperature, power-quality, storage, transport, and platform integration limits.	Needed for BIT thresholds, derating logic, and qualification test coverage.